

Interacting Multipoles of Rank 2 and 2:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \end{array} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ &+3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{aligned} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{aligned} &+105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ &-15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ &-15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ &-15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ &+3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ &+3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \\ &+3 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{aligned} \right] \quad (2b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\delta} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\gamma} \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\beta\delta} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\gamma} \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\delta} \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\delta} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{array} \right] \quad (2c)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Theta_{zz} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \quad (3b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -30 \cdot R_z^4 \cdot \Theta_{z\beta} \cdot \Theta_{z\beta} \\ -30 \cdot R_z^4 \cdot \Theta_{z\alpha} \cdot \Theta_{z\alpha} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta} \cdot \Theta_{\alpha\beta} \end{bmatrix} \quad (3c)$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +105 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -30 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -30 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\gamma\gamma} \\ -15 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{\alpha\alpha} \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\gamma\gamma} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha} \cdot \Theta_{\alpha\alpha} \end{bmatrix} \quad (5)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +45 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +6 \cdot R_z^4 \cdot \Theta_{yy} \cdot \Theta_{yy} \\ +6 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \end{bmatrix} \quad (6a)$$

$$\xRightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \begin{bmatrix} +51 \cdot R_z^4 \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +6 \cdot R_z^4 \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +6 \cdot R_z^4 \cdot \Theta_{yy} \cdot \Theta_{yy} \end{bmatrix} \quad (6b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta}\Theta_{\alpha\beta}\Theta_{\gamma\delta} = \begin{bmatrix} +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz} \cdot \Theta_{zz} \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx} \cdot \Theta_{xx} \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy} \cdot \Theta_{yy} \end{bmatrix} \quad (7)$$